

Chapter 5

Estimation

Introduction

In this chapter, we will learn methods for estimating parameters instead of guessing.

- Method of Moments estimation
- Maximum Likelihood estimation

5.1

Method of Moments Estimators (MMEs)

5.1.1

The idea

Background

Method of moments estimators are constructed by:

- Equating the moments of a population's distribution with the sample moments.
- Solving for the unknown parameters.
- Use as many moments as you need in order to solve for all parameters.

Typically, we start with the lower moments first.

- i.e., $k = 1$ first, then $k = 2$ if necessary, then $k = 3$ if still necessary, etc.

Definition 5.1.1

The k th population (distribution) moment of X , for $k = 1, 2, \dots$, is denoted by μ_k , and defined as

$$\mu_k := E[X^k].$$

Note that $\mu_1 = E[X^1] = \mu$.

Definition 5.1.2

The k th sample moment of X , for $k = 1, 2, \dots$, is denoted by M_k , and defined as

$$M_k := \frac{1}{n} \sum_{i=1}^n X_i^k.$$

Note that $M_1 = \bar{X}$.

5.1.2

Examples

Example 5.1.1

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\lambda)$.

Find an MME of λ .

Example 5.1.1 (cont)

Example 5.1.2

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Gamma}(\alpha, \beta)$.

Find MMEs of α and β .

Example 5.1.2 (cont)

Example 5.1.2 (cont)

Example 5.1.2 (cont)

5.1.3

Properties of MMEs

Pros of MMEs

- Intuitive
- Easy to find, overall.
- Usually consistent.
- Usually asymptotically normal.

Cons of MMEs

- Often biased.
- Not uniquely defined.
- May be outside of the parameter space.
- May not exist.

Example 5.1.1 continued

In Example 5.1.1, we derived the MME $\hat{\lambda} = \bar{X}$ for an Exponential λ population.

What can we say about the convergence of the estimator?

Example 5.1.1 (cont)

Example 5.1.1 (cont)

Example 5.1.2 continued

In Example 5.1.2, we derived the MME estimator

$$\hat{\alpha} = \frac{\bar{X}^2}{\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2} = \frac{M_1^2}{M_2 - M_1^2}.$$

Example 5.1.2 (cont)

Example 5.1.2 (cont)

Some thoughts

- The WLLN gives us that $M_k \xrightarrow{P} \mu_k$ as long as $\text{var}(X^k)$ is finite.
- Asymptotic normality using the CLT and combining the WLLN with the Delta Method.
- MMEs are not well-defined. We could use the second moments to derive an estimator instead of the first.
- MMEs may take values that fall outside the parameter space.
- MMEs may not exist if moments don't exist (i.e., aren't finite).

5.2

Maximum Likelihood Estimators

Idea

A maximum likelihood estimator (MLE) estimates an unknown parameter with the value whose associated model is most likely to have produced the observed data.

The joint density

In general, for fixed $\vec{x} = (x_1, x_2, \dots, x_n)$, the joint density is defined as

$$f(\vec{x}; \theta) = f(x_1, x_2, \dots, x_n; \theta).$$

Note that θ is assumed fixed in the density function since its value is conditional on θ .

Joint density versus likelihood function

Mathematically, the joint density and the likelihood function are identical.

Conceptually, the likelihood function differs from the joint density in that θ is assumed to be unknown and the data, $\vec{x} = (x_1, x_2, \dots, x_n)$ are fixed.

The likelihood function

The likelihood function is denoted

$$L(\theta) = L(\theta; \vec{x}) = f(\vec{x}; \theta).$$

Note that θ is assumed fixed in the density function since its value is conditional on θ .

Definition 5.2.1

Suppose that $X_1, X_2, \dots, X_n$ have joint pdf $f(\vec{x}; \theta)$.

A **likelihood function** is denoted by $L(\theta)$ is defined as any function proportional to $f(\vec{x}; \theta)$ when thought of as a function of θ .

The **maximum likelihood estimator** of θ is the value of θ that maximizes $L(\theta)$.

5.2.1

Examples

Example 5.2.2

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$. Determine the MLE of p .

Example 5.2.2 (cont)

Example 5.2.2 (cont)

Notation

Given a likelihood function $L(\theta)$, we denote the log-likelihood function by

$$\ell(\theta) = \ln L(\theta).$$

The maximum likelihood estimator for θ that maximizes $L(\theta)$, and equivalently, $\ell(\theta)$, since $\ln$ is an order-preserving transformation.

Example 5.2.3

Let $X_1, X_2, \dots, X_n$ be a random sample with pdf

$$f(x; \theta) = \theta x^{\theta-1} I_{(0, \infty)}(x), \quad \text{for } \theta > 0.$$

Determine the MLE of θ .

Example 5.2.3 (cont)

Example 5.2.3 (cont)

Example 5.2.4

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$. Determine the MLEs for μ and σ^2 .

Example 5.2.4 (cont)

Example 5.2.4 (cont)

Example 5.2.4 (cont)

Example 5.2.4 (cont)

Example 5.2.5 (Parameters in the Support of the Distribution)

Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution with pdf

$$f(\vec{x}; \theta, \lambda) = \theta \lambda^\theta x^{-(\theta+1)} I_{(\lambda, \infty)}(x),$$

with $\theta > 0, \lambda > 0$. Find the MLEs for θ and λ .

Example 5.2.5 (cont)

Example 5.2.5 (cont)

Example 5.2.5 (cont)

Example 5.2.5 (cont)

Example 5.2.5 (cont)

5.2.2

The Invariance Property of MLEs

Definition 5.2.2

If $\hat{\theta}$ is the MLE of θ , then the MLE of a function $\tau(\theta)$ is

$$\widehat{\tau(\theta)} = \tau(\hat{\theta}).$$

This is known as the **invariance property** of MLEs.

Invariance examples

Invariance examples (cont)

Pseudo proof of invariance

Pseudo proof of invariance (cont)

5.3

The Cramér–Rao Lower Bound (CRLB)

CRLB Introduction

The CRLB is the minimum possible variance of an unbiased estimator of a parameter θ (or a function, $\tau(\theta)$).

If $\hat{\theta}$ is an unbiased estimator of θ , then the CRLB tells us how small the variance of $\hat{\theta}$ can be.

- This lower bound may not be attainable.
- If an unbiased estimator does attain the associated CRLB, then it has the uniformly minimum variance among all unbiased estimators.

5.3.1

Statement of the CRLB and Derivation

The Cramér–Rao Lower Bound

Let $X_1, X_2, \dots, X_n$ be random variable with joint pdf $f(\vec{x}; \theta)$. Assume that the support of the distribution does not depend on θ and that θ is one-dimensional.

Consider estimating some function $\tau(\theta)$ and let $T = t(\vec{X})$ be any unbiased estimator of $\tau(\theta)$.

Then

$$\text{var}[T] \geq \frac{[\tau'(\theta)]^2}{E \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right]},$$

under certain regularity assumptions.

Regularity assumptions for the CRLB

The regularity assumption for the CRLB are:

- $0 < E \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right) \right] < \infty,$
- $\frac{\partial}{\partial \theta} \int f(\vec{x}; \theta) d \vec{x} = \int \frac{\partial}{\partial \theta} f(\vec{x}; \theta) d \vec{x}$
- $\frac{\partial}{\partial \theta} \ln f(\vec{x}; \theta)$ exists.

CRLB notation

We denote the CRLB by

$$\text{CRLB}_{\tau(\theta)} := \frac{[\tau'(\theta)]^2}{E \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right]}.$$

Computation of the CRLB will simplify when the data are i.i.d., but the CRLB holds for more general joint pdfs.

Definition 5.3.1

The Fisher information for a sample is defined as

$$I_n(\theta) = E \left[\left(\frac{\partial}{\partial \theta} \ln f(\vec{X}; \theta) \right)^2 \right].$$

The Fisher information quantifies the amount of information the sample contains about θ .

We can rewrite the CRLB of θ as

$$\text{CRLB}_{\tau(\theta)} = \frac{[\tau'(\theta)]^2}{I_n(\theta)}.$$

The Cauchy-Schwarz Inequality

For square-integrable functions f and g ,

$$\left(\int f(x)g(x) \, dx \right)^2 \leq \left(\int f(x)^2 \, dx \right) \left(\int g(x)^2 \, dx \right)$$

There are analagous Cauchy-Schwarz inequalities for sums and inner products.

Proof of C-S inequality

Proof of C-S inequality (cont)

Proof of C-S inequality (cont)

Proof of CRLB

Proof of CRLB (cont)

Proof of Equation 5.3.4

Proof of Equation 5.3.4 (cont)

Proof of Equation 5.3.4 (cont)

Proof of Equation 5.3.4 (cont)

Example 5.3.1

Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$.

- a. Find the CRLB for the variance of all unbiased estimators of p .
- b. Find the CRLB for the variance of all unbiased estimators of $\tau(p) = p(1 - p)$.

Example 5.3.1 (cont)

Example 5.3.1 (cont)

Example 5.3.1 (cont)

5.3.2

Computational Simplifications for the CRLB